AI CAMPUS ASSISTANT(RAG-BASED)

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ABSTRACT

The AI Campus Assistant is an intelligent chatbot designed to help students quickly find information from campus documents such as fee structure, syllabus, and admission rules. Traditional chatbots use predefined responses and cannot understand document content. This system uses Retrieval-Augmented Generation (RAG) to read documents and retrieve relevant information. The text is converted into embeddings and stored in a vector database. When a student asks a question, the system retrieves related document sections and generates a response using an AI language model. It also supports multilingual answers, voice queries, and chat memory. The system improves communication and reduces the workload of administrative staff.

INTRODUCTION

Educational institutions provide large amounts of information through documents such as notices, PDFs, and academic guidelines. Students often find it difficult to quickly locate the required information. Traditional chatbots rely on predefined responses and cannot read updated documents. The AI Campus Assistant solves this problem using AI and Retrieval-Augmented Generation. The system retrieves relevant document content and generates accurate responses. It supports features like voice input, multilingual responses, and chat history. This improves accessibility and helps students get information quickly.

PROBLEM STATEMENT

Students frequently need information about admissions, fees, syllabus, and examination rules. This information is usually available in large PDF documents, making it difficult to search quickly. Traditional campus chatbots cannot dynamically read or understand these documents. As a result, students repeatedly contact administrative staff for information. This increases workload and delays communication. Therefore, an intelligent system is needed to automatically retrieve document information and answer queries accurately. The AI Campus Assistant addresses this problem using AI-based document retrieval.

OBJECTIVES

The main objective of the project is to develop an AI-based campus assistant that can answer student queries using institutional documents. The system reads and understands documents such as admission rules, syllabus, and fee structures. It retrieves relevant information and generates accurate responses. The project also aims to reduce administrative workload. Additional objectives include supporting multiple document formats, multilingual responses, voice queries, and chat history memory. The system also provides an admin dashboard for document management.

PROJECT REQUIREMENTS

HARDWARE

- Processor : Intel i5
- RAM : 8 gb
- Storage : 500gb
- Microphone(for voice input)

SOFTWARE

- OS: Windows
- Python 3.10.11
- Flask
- LangChain
- FAISS
- PyPDF
- Speechrecognition library

MODULES

- The Document Loader Module extracts text from uploaded documents.
- The Vector Database Module converts text into embeddings and stores them for search.
- The Query Processing Module receives user questions.
- The Retrieval Module finds relevant document sections
- The LLM Integration Module generates responses using AI models.
- Additional modules include Multilingual Translation, Chat History Memory, Admin Document Management, and Voice-to-Query.

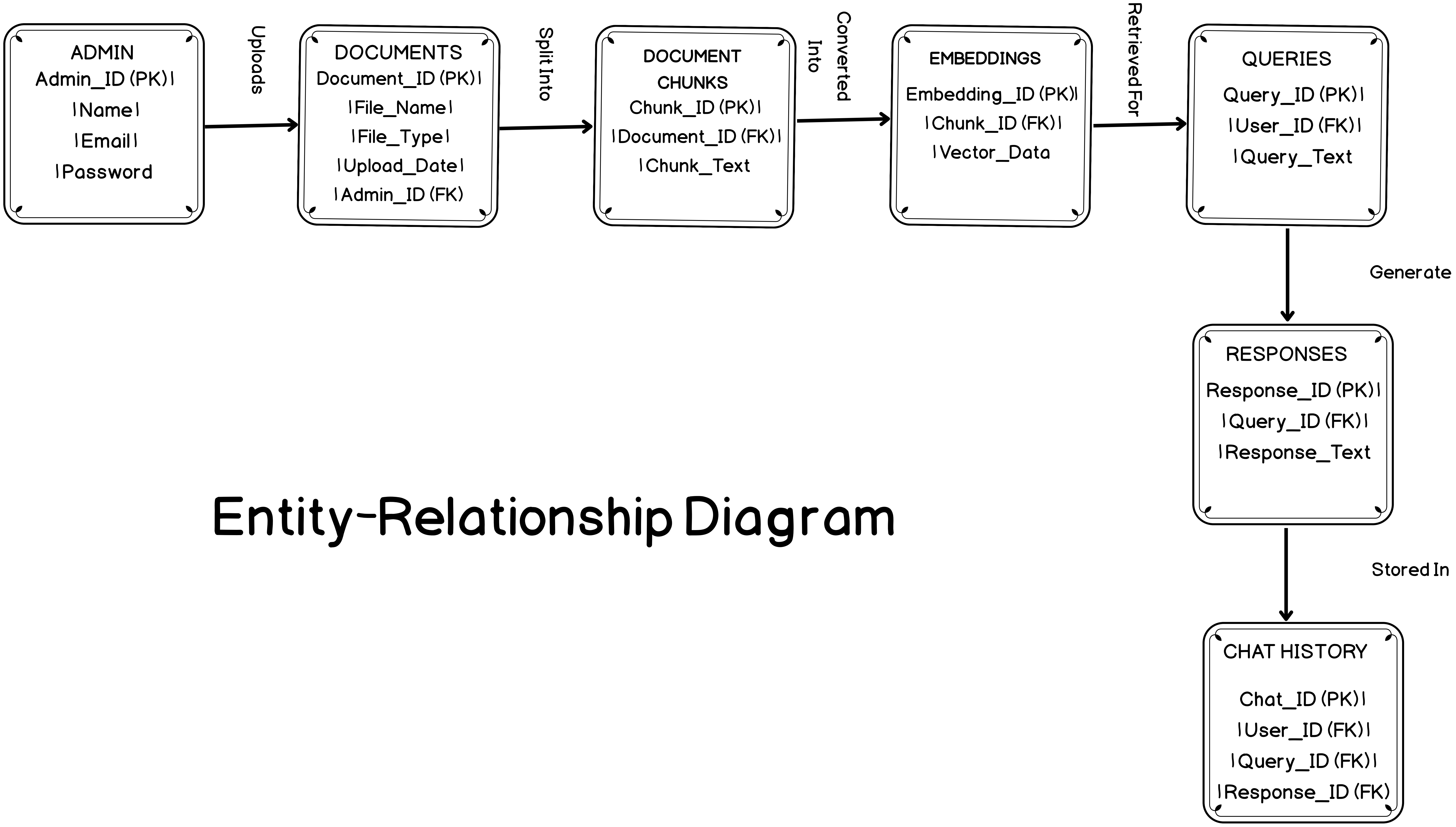
METHODOLOGY

The system begins by uploading campus documents through the admin dashboard. The system extracts text and divides it into smaller chunks. Each chunk is converted into embeddings and stored in a vector database. When a user asks a question, the query is also converted into an embedding. The system retrieves the most relevant document sections using similarity search. These sections are sent to an AI language model. The model generates a response which is displayed to the user.

ADVANTAGE

ADVANTAGE

- Provides accurate, document-based answers
- Reduces administrative workload
- Supports multilingual and voice interaction
- Easy to update documents without code changes
- Improves student accessibility and satisfaction



Entity-Relationship Diagram

DFD Level 0

```
+-------------------+
|     ADMIN     |
+-------------------+
        |
|Upload / Delete Documents
        v
+-------------------------+
| AI Campus Assistant  |
|     System       |
+-------------------------+
        ^
        |
   |Ask Question
+-------------------+
|    STUDENT    |
+-------------------+
        |
   |Response
        v
+-------------------+
|    STUDENT    |
+-------------------+
```

DATA FLOW DIAGRAM(DFD)

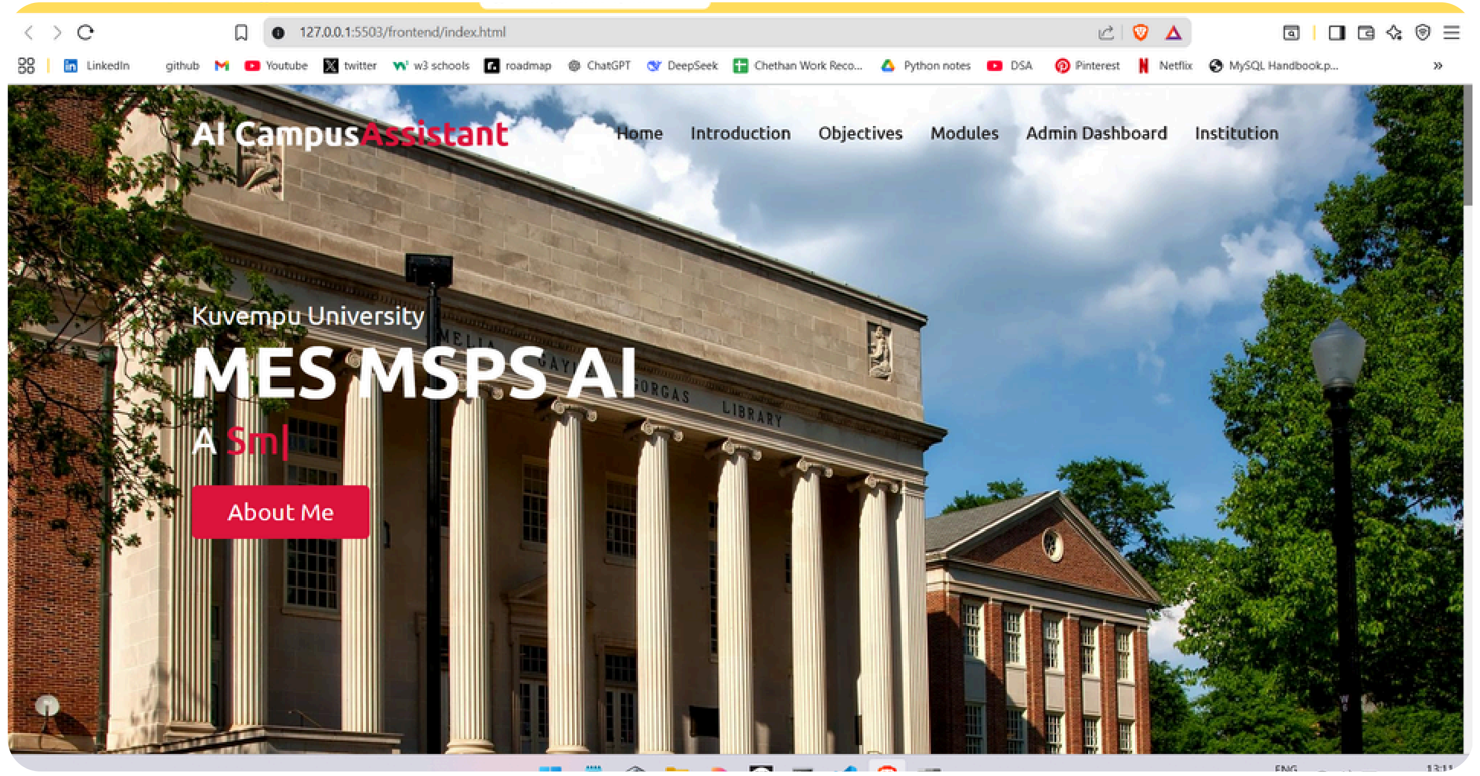
DFD Level 1 (System Modules)

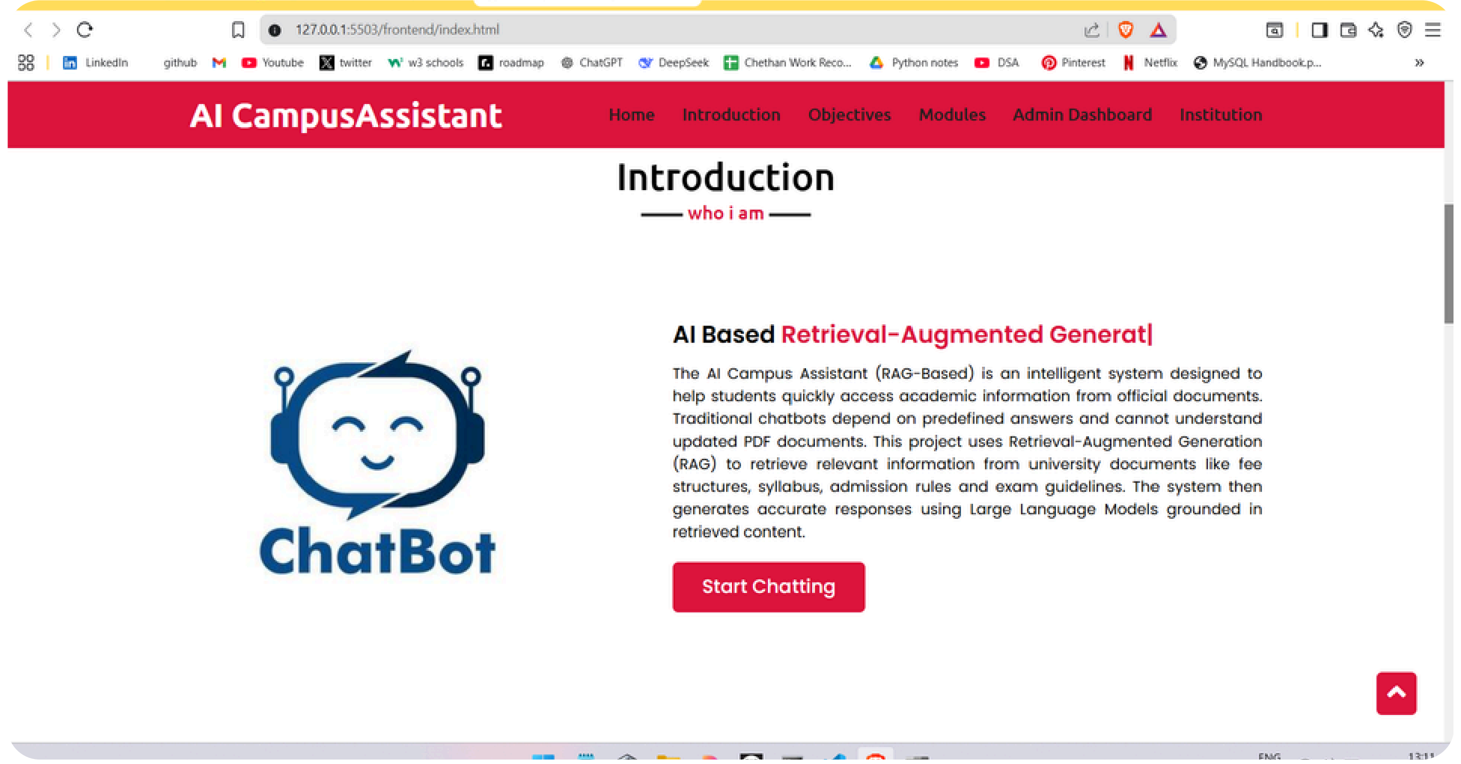
```
       Student
          |
       |Query
          v
+--------------------+
|Query Processing  |
+--------------------+
          |
          v
+--------------------+
|Query Embedding  |
+--------------------+
          |
          v
+--------------------+
|Vector Search   |
|(FAISS Database)  |
+--------------------+
          |
          v
+--------------------+
|Context Retrieval |
+--------------------+
          |
          v
+--------------------+
|LLM Response Engine|
+--------------------+
          |
          v
+--------------------+
|Response Delivery  |
+--------------------+
```

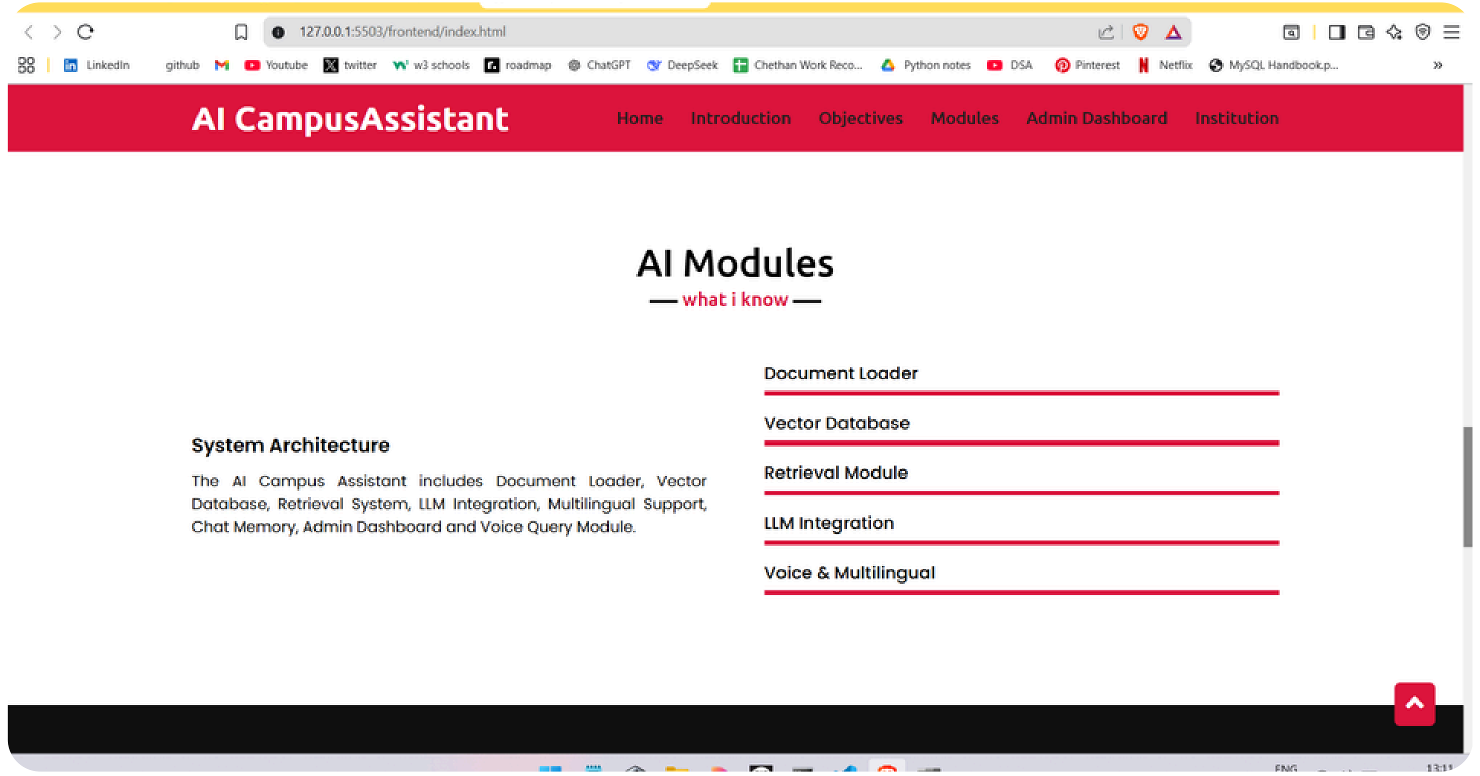
Admin Data Flow (Document Processing)

```
        Admin
          |
   |Upload Document
          v
+---------------------+
|Document Loader   |
+---------------------+
          |
          v
+---------------------+
|Text Extraction   |
|(PDF/DOCX Parser) |
+---------------------+
          |
          v
+---------------------+
|Text Chunking    |
+---------------------+
          |
          v
+---------------------+
|Embedding Creation |
+---------------------+
          |
          v
+---------------------+
|FAISS Vector DB   |
+---------------------+
```

SNAPSHOTS


127.0.0.1:5503/frontend/index.html
AI CampusAssistant
Home
Introduction
Objectives
Modules
Admin Dashboard
Institution
Kuvempu University
MES MSPS AI
A Sm|
About Me


127.0.0.1:5503/frontend/index.html
AI CampusAssistant
Home
Introduction
Objectives
Modules
Admin Dashboard
Institution
Introduction
who i am
ChatBot
AI Based Retrieval-Augmented Generat|
The AI Campus Assistant (RAG-Based) is an intelligent system designed to help students quickly access academic information from official documents. Traditional chatbots depend on predefined answers and cannot understand updated PDF documents. This project uses Retrieval-Augmented Generation (RAG) to retrieve relevant information from university documents like fee structures, syllabus, admission rules and exam guidelines. The system then generates accurate responses using Large Language Models grounded in retrieved content.
Start Chatting


127.0.0.1:5503/frontend/index.html
AI CampusAssistant
Home
Introduction
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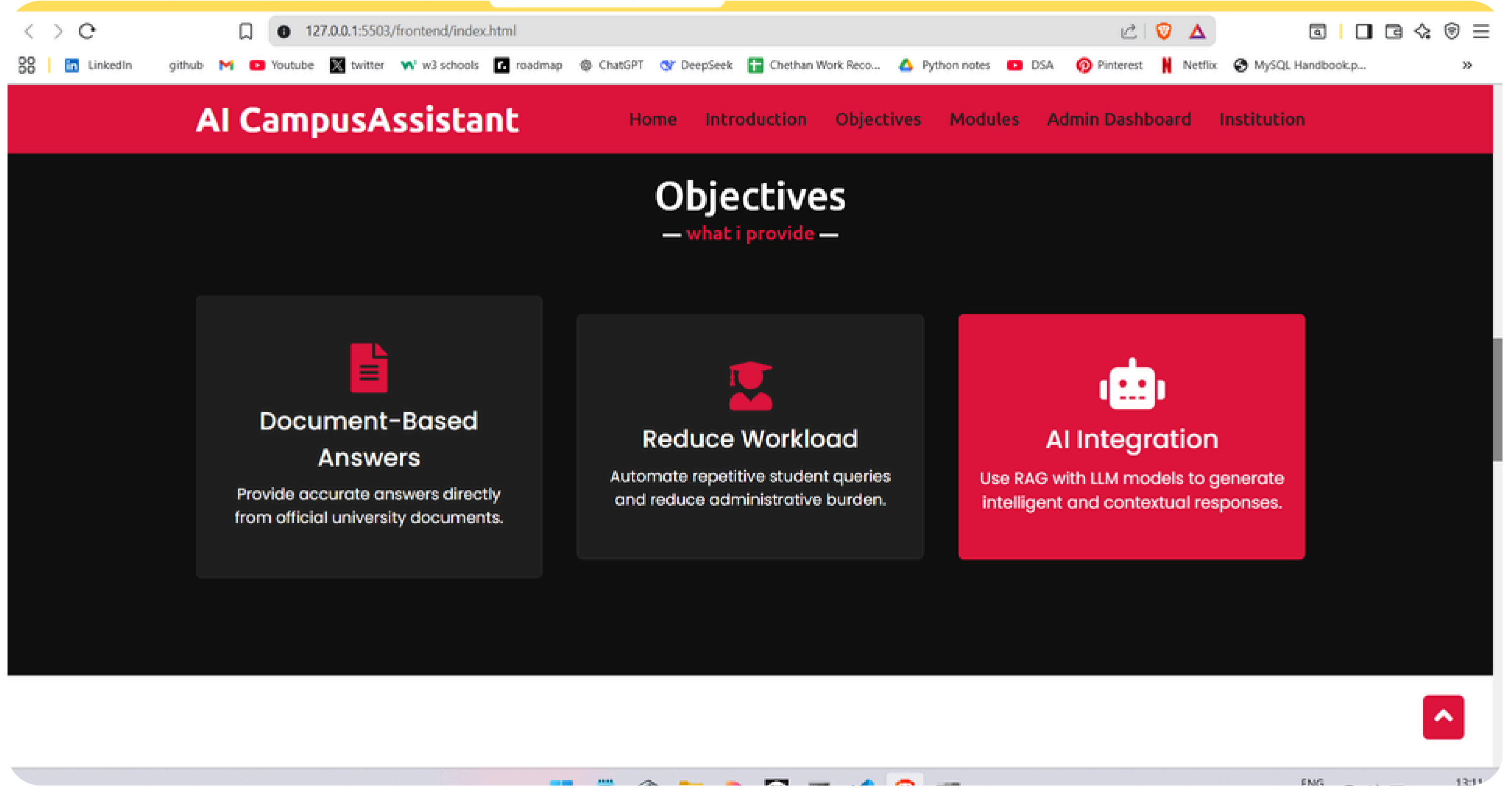
Admin Dashboard
Institution
AI Modules
what i know
System Architecture
The AI Campus Assistant includes Document Loader, Vector Database, Retrieval System, LLM Integration, Multilingual Support, Chat Memory, Admin Dashboard and Voice Query Module.
Document Loader
Vector Database
Retrieval Module
LLM Integration
Voice & Multilingual


127.0.0.1:5503/frontend/index.html
AI CampusAssistant
Home
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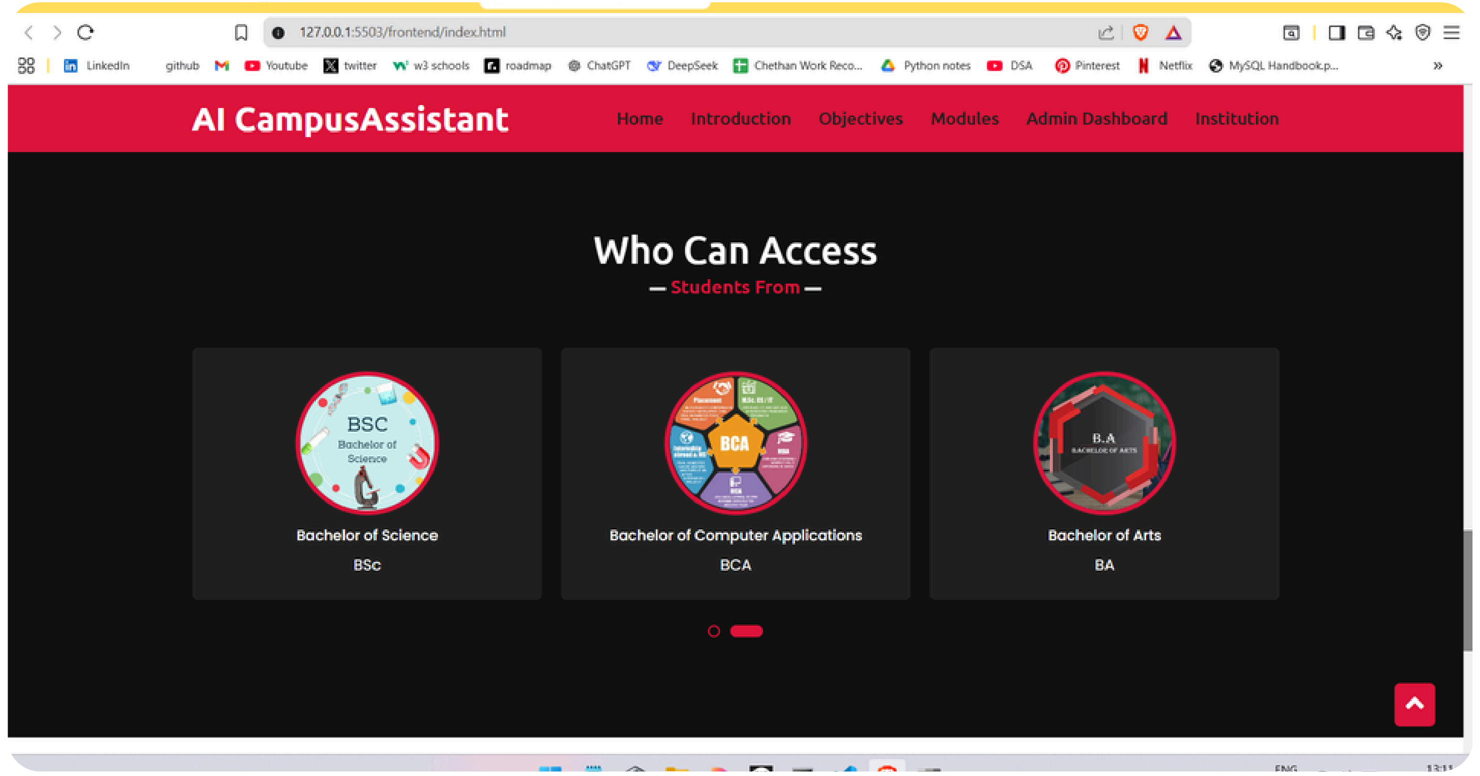
Institution
Objectives
what i provide
Document-Based Answers
Provide accurate answers directly from official university documents.
Reduce Workload
Automate repetitive student queries and reduce administrative burden.
AI Integration
Use RAG with LLM models to generate intelligent and contextual responses.

SNAPSHOTS



AI CampusAssistant
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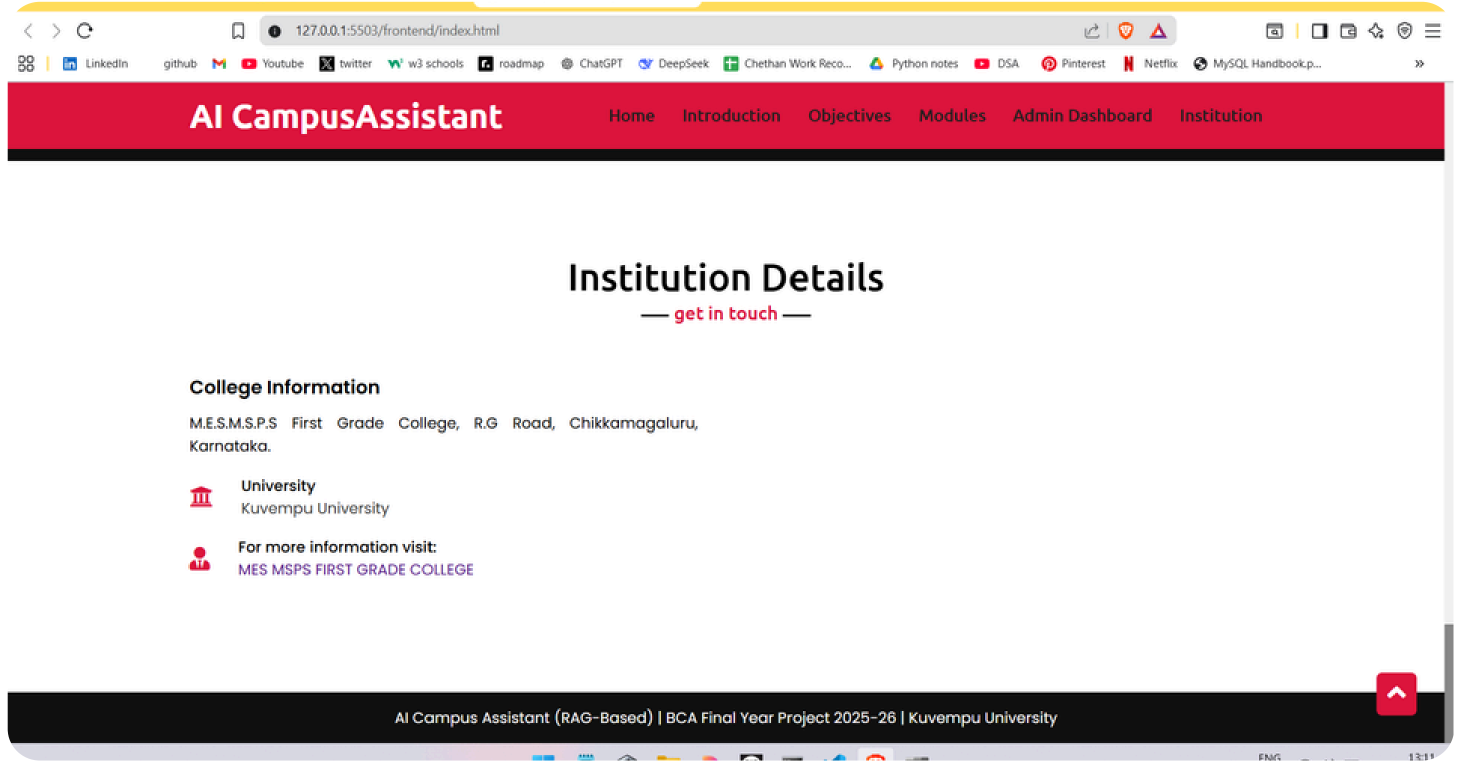
Institution
Who Can Access
— Students From —
Bachelor of Science
BSc
Bachelor of Computer Applications
BCA
Bachelor of Arts
BA



AI CampusAssistant
Home
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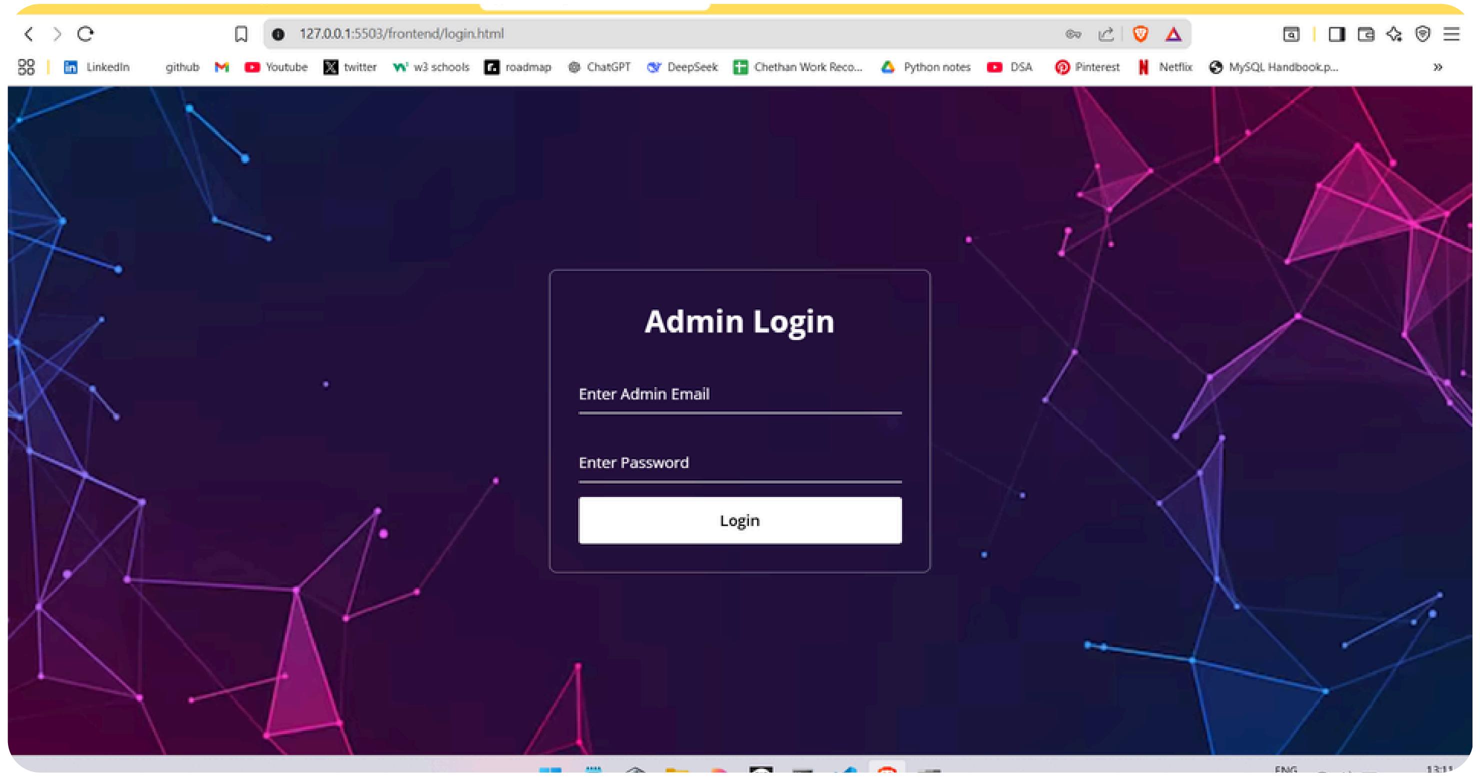
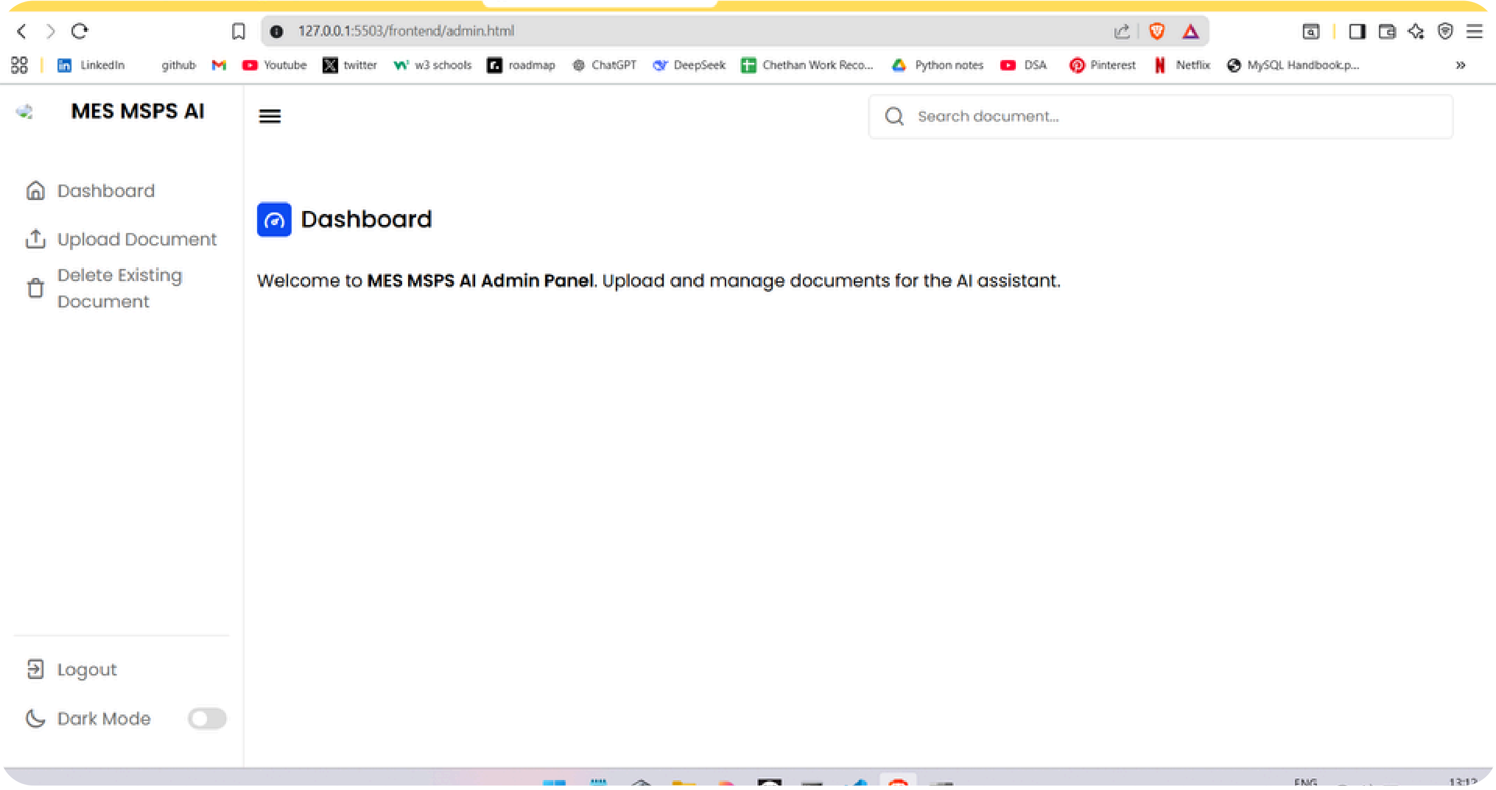
Admin Dashboard
Institution
Institution Details
— get in touch —
College Information
M.E.S.M.S.P.S First Grade College, R.G Road, Chikkamagaluru, Karnataka.
University
Kuvempu University
For more information visit:
MES MSPS FIRST GRADE COLLEGE
AI Campus Assistant (RAG-Based) | BCA Final Year Project 2025-26 | Kuvempu University

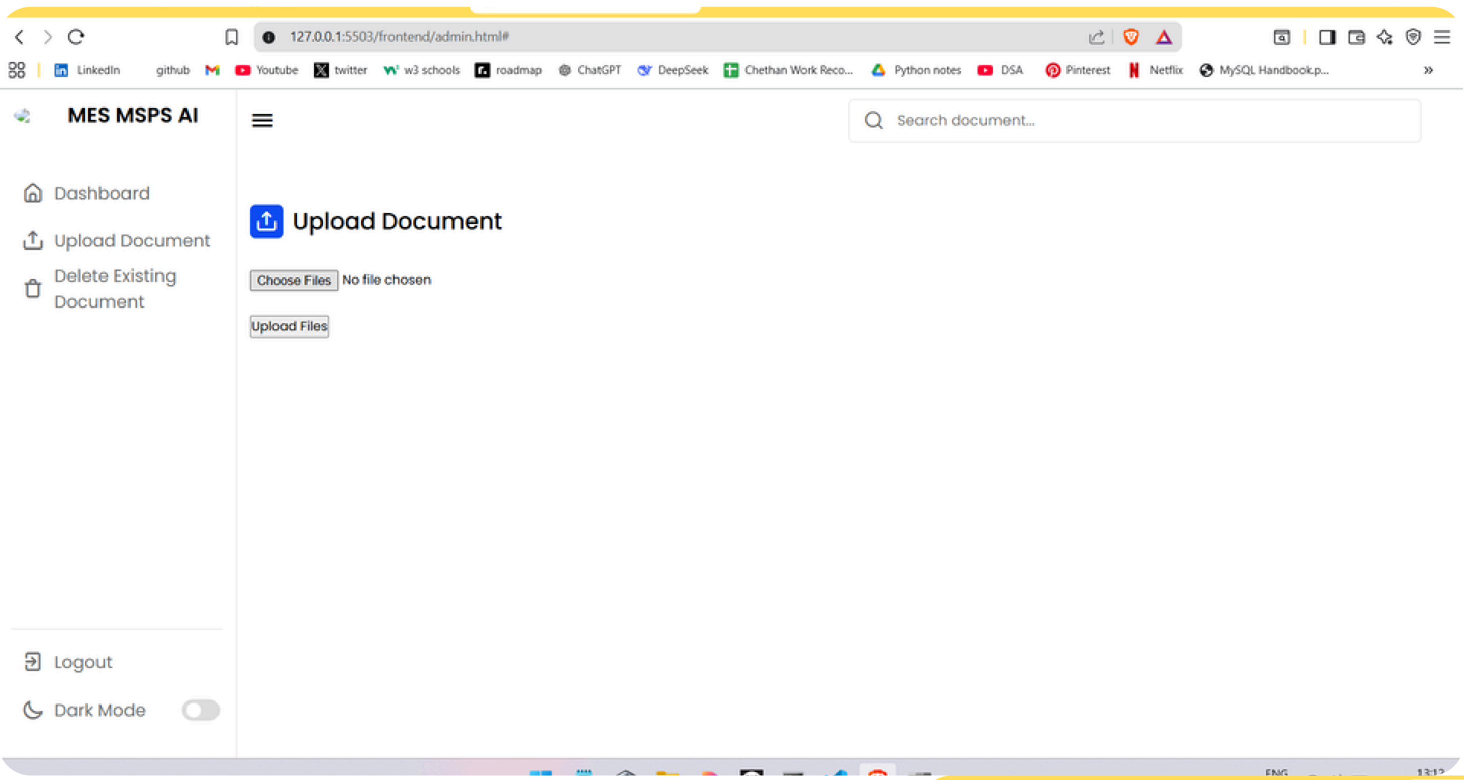


Admin Login
Enter Admin Email
Enter Password
Login

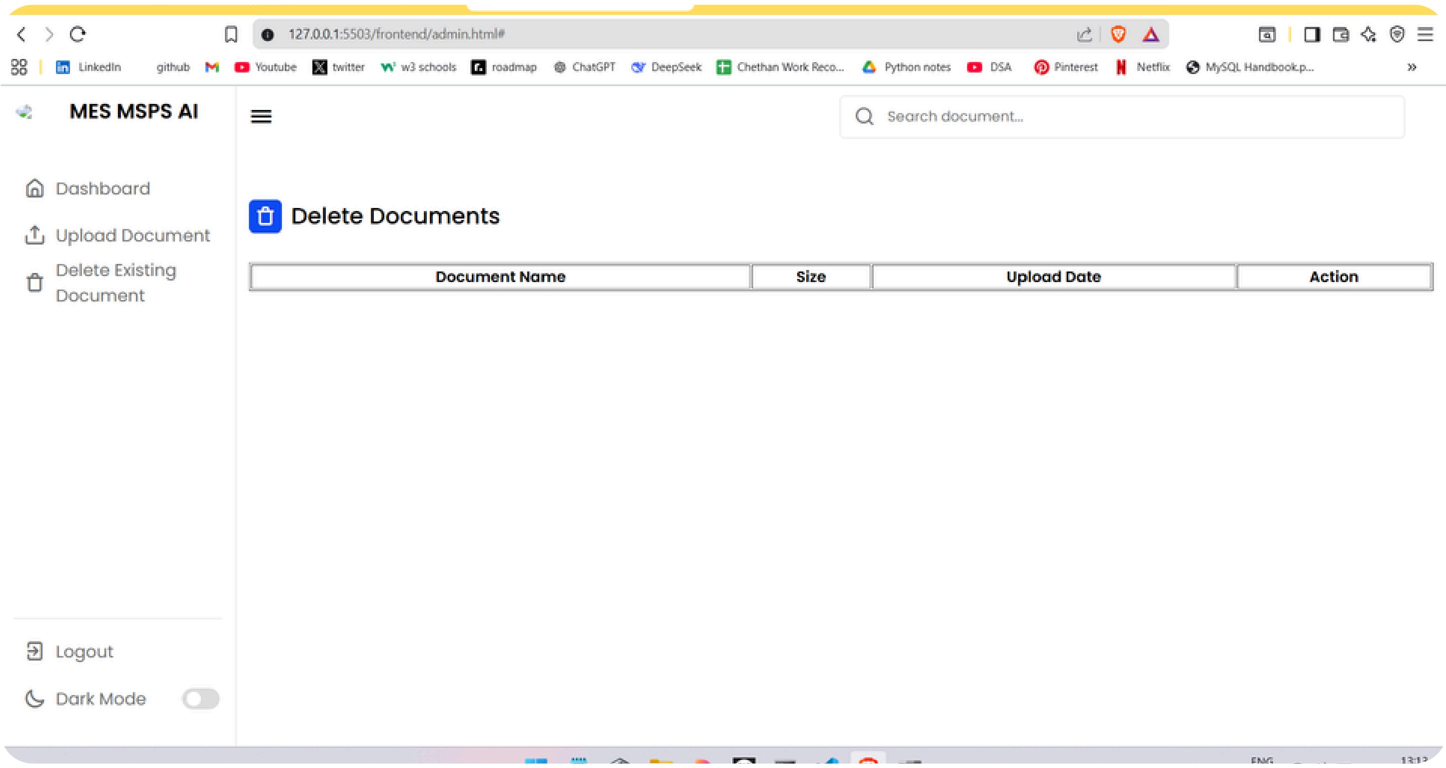


MES MSPS AI
Search document...
Dashboard
Upload Document
Delete Existing Document
Dashboard
Welcome to MES MSPS AI Admin Panel. Upload and manage documents for the AI assistant.
Logout
Dark Mode

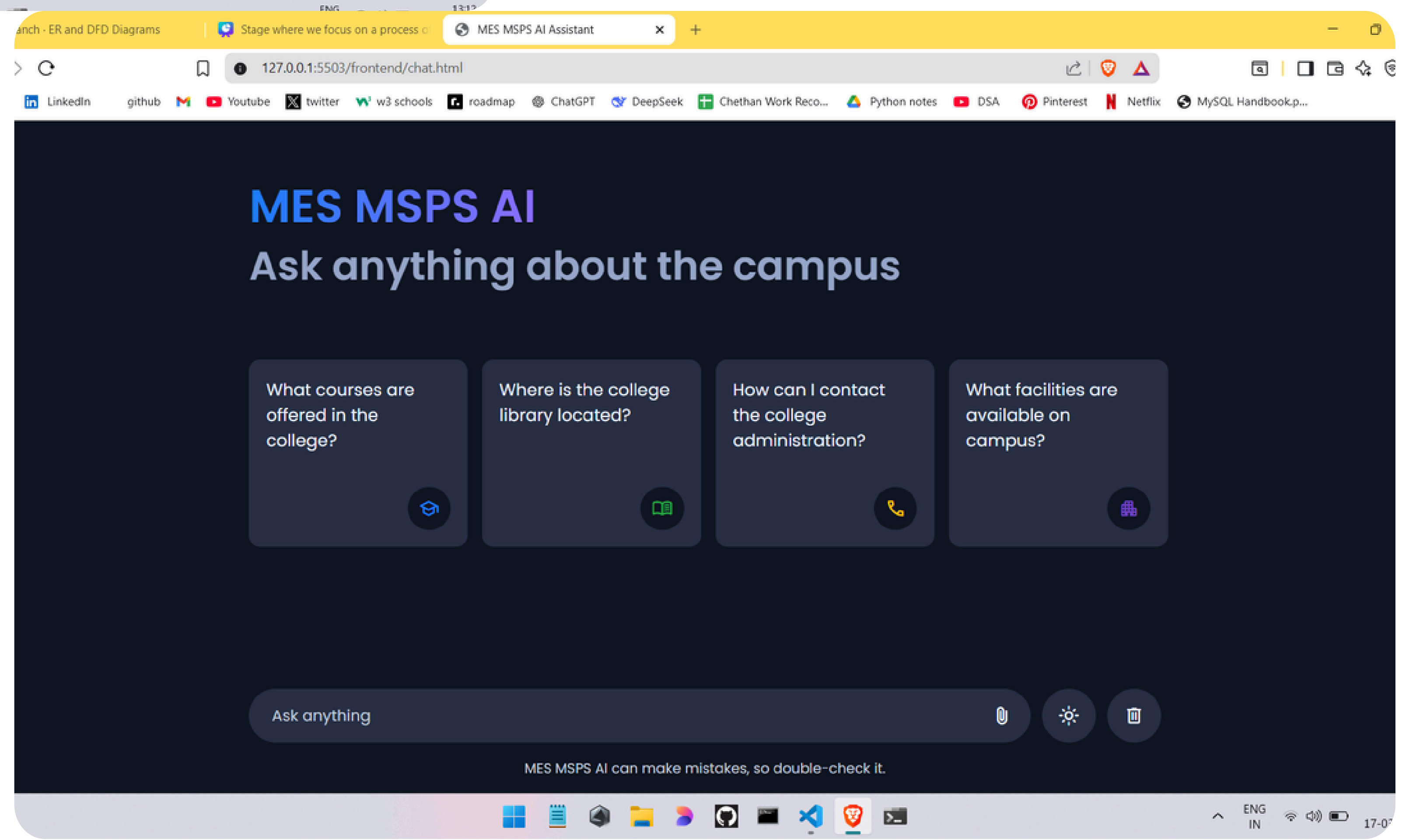
SNAPSHOTS



127.0.0.1:5503/frontend/admin.html#
MES MSPS AI
Search document...
Dashboard
Upload Document
Delete Existing Document
Upload Document
Choose Files No file chosen
Upload Files
Logout
Dark Mode



127.0.0.1:5503/frontend/admin.html#
MES MSPS AI
Search document...
Dashboard
Upload Document
Delete Existing Document
Delete Documents
Document Name
Size
Upload Date
Action
Logout
Dark Mode



MES MSPS AI Assistant
127.0.0.1:5503/frontend/chat.html
MES MSPS AI
Ask anything about the campus
What courses are offered in the college?
Where is the college library located?
How can I contact the college administration?
What facilities are available on campus?
Ask anything
MES MSPS AI can make mistakes, so double-check it.

CONCLUSION

The AI Campus Assistant (RAG-Based) provides an effective solution to the limitations of traditional campus chatbots. By combining document retrieval with AI-based response generation, the system ensures that student queries are answered accurately and reliably using official university documents. The inclusion of advanced features such as multilingual support, conversational memory, admin document control, and voice-based queries makes the system suitable for real-world campus environments. This project demonstrates the practical application of modern AI technologies in the education sector and highlights how intelligent systems can improve communication, efficiency, and accessibility. The proposed solution is scalable, user-friendly, and well-suited for deployment in academic institutions.

THANK YOU